

Problem 1 :

Is set of odd numbers with binary operations  $(+)$ , i.e.,  $\langle O, + \rangle$  an abelian group? If not explain the reasons with necessary notations.

Step 1: Recall group axioms

A set  $G$  with a binary operation  $*$  is a group if it satisfies:

1. Closure: For all  $a, b \in G$ ,  $a * b \in G$ .
2. Associativity:  $(a * b) * c = a * (b * c)$ .
3. Identity element: There exists  $e \in G$  such that  $a * e = e * a = a$  for all  $a \in G$ .
4. Inverse element: For each  $a \in G$ , there exists  $a^{-1} \in G$  such that  $a * a^{-1} = e$ .

Additionally, if  $a * b = b * a$  for all  $a, b \in G$  then the group is abelian.

Step 2: Take the set of odd integers

Let

$$O = \{\dots, -3, -1, 1, 3, 5, \dots\}$$

with binary operation  $+$  (usual addition).

Step 3: Verify group axioms

1. Closure:

$$\text{Odd} + \text{Odd} = \text{Even}$$

$$\text{Example: } 3 + 5 = 8 \text{ (not odd)}$$

Thus, closure fails.

Already disqualified from being a group.

2. Associativity:

Addition of integers is associative:

$$(a+b)+c = a+(b+c)$$

Holds (but irrelevant since closure failed)

### 3. Identity element:

The additive identity in  $(\mathbb{Z}, +)$  is 0.

But 0 is not odd, so 0 has no identity element

× Fails.

### 4. Inverse element:

For an odd integer  $a$ , its inverse under addition is  $-a$ . Example: If  $3 \in O$ , inverse is  $-3$ , which is also odd.

This works.

### Step 4: Conclusion

Since closure and identity fail, the set of odd integers with  $+$  is not a group. Therefore, it cannot be an abelian group either.

Final Answer:

The set of odd integers under addition is not an abelian group because:

\*It is not closed ( $\text{odd} + \text{odd} = \text{even} \notin \text{odd}$ ).

\*It does not contain the identity element (since 0 is not odd).

2.